Example 3: Compare 2456 and 3467.

Solution: Both are 4-digit numbers
So we compare first extreme left different digits
which are 2 and 3

Here $2 < 3$

So, $2456 < 3467$

Activity **Compare 67835 and 67541**

Solution: Both are ☐ digit numbers

We compare first extreme left different digits

which are ☐ and ☐

Here 8 ☐ 5

So, 67835 ☐ 67541

EXERCISE 5

(1) Fill in the blanks by using symbols <, > or =.

(i)	**245**	☐	**3167**	(ii)	**54231**	☐	**964**
(iii)	**3105**	☐	**3105**	(iv)	**5421**	☐	**5418**
(v)	**672315**	☐	**871237**	(vi)	**120001**	☐	**105552**

(2) Compare the numbers by using symbols <, > or =.

(i)	**542**	and	**6712**	(ii)	**94562**	and	**4392**
(iii)	**6324**	and	**6324**	(iv)	**6421**	and	**6578**
(v)	**94561**	and	**94271**	(vi)	**345671**	and	**345921**

Write the given set of numbers in ascending and descending order.

Starter: The teacher will write the following numbers on the board in random order:

35, 12, 30

The teacher will then ask one student to come to the board and write these numbers from the smallest to the largest:

12, 30, 35

Explain that this is called ascending order.

The teacher will choose another student to come to the board and write the same numbers from largest to smallest:

35, 30, 12

Explain that this is called descending order .

The teacher can do a few more exercises choosing different children each time, to reinforce this vocabulary.

EXERCISE 6

1. Write the following numbers in ascending order in given boxes.

(i) 476, 52341, 7881, 5034 ____, ____, ____, ____,

(ii) 2346, 4632, 2354, 1778 ____, ____, ____, ____,

(iii) 43513, 3451, 53314, 41353 ____, ____, ____, ____,

2. Write the following numbers in descending orders in given boxes.

(i) 6432, 3213, 4343, 7120 ____, ____, ____, ____,

(ii) 49231, 12349, 94321, 31249 ____, ____, ____, ____,

(iii) 120451, 12345, 57401, 10000 ____, ____, ____, ____,

1.6 NUMBER LINE

Represent and identify the value of given number on number line

Starter: On the board, the teacher will draw a line and mark numbers at equal distances, from the least to the greatest.
Emphases that the intervals must be of equal distance.
For example: from 0 to 10. Rub out some numbers and ask a students to fill in the missing numbers.

Draw another number line with different set of numbers.
For example: 10 to 20. Rub some numbers and ask a student to fill in the missing numbers.

A number can be represented by taking equal distance on a line

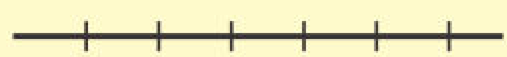
The line on which numbers are represented is called **Number Line.**

A number line

Example : Represent 4 on the number line.

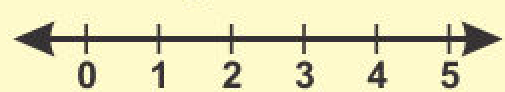
Step 1: Draw a line with scale.

Step 2: Mark points at equal distances.



Step 3: Select a point from left to right.

Step 4: Represent numbers in ascending order from starting point.

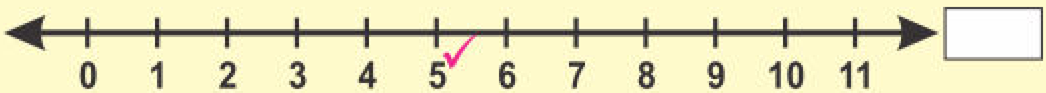


Step 5: Identify the given number 4.

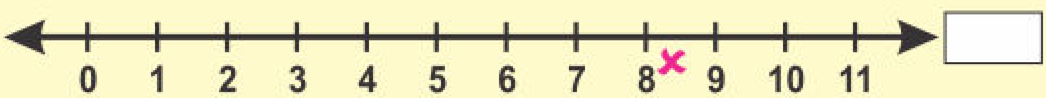
Teacher's Note

Teacher should help the students to identify the value of numbers from the number line.

Activity Which number has been ticked (✓) at the number line.



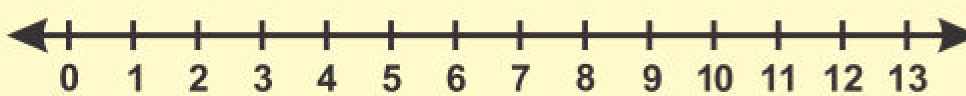
Activity Which number has been crossed (×) at the number line.



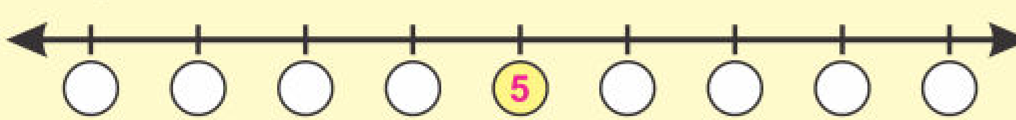
EXERCISE 7

1. Represent 6 on a number line.

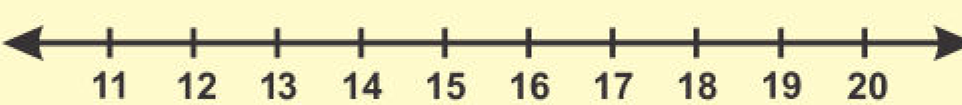
2. Represent the following numbers on given number line.
6, 7, 9, 10 and 12.



3. Fill in the blanks with before and after numbers on the given number line.



4. Encircle the following numbers on the given number line.
14, 16 and 18.



Unit 2

NUMBER OPERATIONS

2.1 ADDITION

We have already learnt how to add numbers up to 3-digit in previous class.

EXERCISE 8

Solve.

1. $\begin{array}{r} 4\ 3 \\ +\ 2\ 4 \\ \hline \end{array}$

2. $\begin{array}{r} 4\ 3\ 1 \\ +\ 2\ 4 \\ \hline \end{array}$

3. $\begin{array}{r} 6\ 5 \\ +\ 2\ 4 \\ \hline \end{array}$

4. $\begin{array}{r} 1\ 1\ 4 \\ +\ 8\ 5 \\ \hline \end{array}$

5. $\begin{array}{r} 5\ 8\ 2 \\ +\ 1\ 3\ 7 \\ \hline \end{array}$

6. $\begin{array}{r} 9\ 0\ 5 \\ +\ 1\ 8\ 2 \\ \hline \end{array}$

7. $\begin{array}{r} 6\ 3\ 1 \\ +\ 2\ 8\ 9 \\ \hline \end{array}$

8. $\begin{array}{r} 5\ 9\ 9 \\ +\ 3\ 2\ 4 \\ \hline \end{array}$

9. $\begin{array}{r} 4\ 8\ 8 \\ +\ 3\ 2\ 5 \\ \hline \end{array}$

10. 27 + 22 = ____

11. 84 + 101 = ____

12. 850 + 88 = ____

13. 104 + 645 = ____

14. 542 + 227 = ____

15. 300 + 200 = ____

Add numbers up to 4-digits (with and without carrying) vertically and horizontally

First we add 4-digit numbers without carrying.

Starter: Using Example 1, the teacher will explain step wise, the vertical method of addition.

To check for student understanding, another sum will be written on the board and individual children will be asked to come to the board and solve the sum step by step.

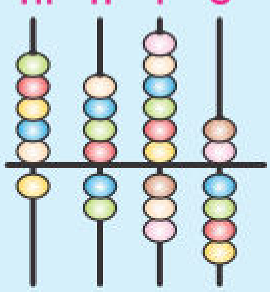
The children will then do Exercise 9

Questions 1- 6

Example 1: **Add 5462 and 1234**

Solution: **(i) Vertical method:**

	Th	H	T	O
	5	4	6	2
+	1	2	3	4
	6	6	9	6



Step 1: Add ones
2 ones + 4 ones = 6 ones

Step 2: Add tens
6 tens + 3 tens = 9 tens

Step 3: Add hundreds
4 hundreds + 2 hundreds = 6 hundreds

Step 4: Now add thousands
5 thousands + 1 thousand = 6 thousands

(ii) Horzontal method:

Or 5462 + 1234 = ☐

Th	H	T	O		Th	H	T	O		Th	H	T	O
5	4	6	2	+	1	2	3	4	=	6	6	9	6

Step 1:	Add the ones	**2 + 4 = 6**
Step 2:	Add the tens	**6 + 3 = 9**
Step 3:	Add the hundreds	**4 + 2 = 6**
Step 4:	Add the thousands	**5 + 1 = 6**

Note: Start from ones. Add ones first, then tens, then hundreds and add thousands in last.

EXERCISE 9

(A) Solve:

(1)
```
  2 5 6 4
+ 1 3 2 1
---------
```

(2)
```
  3 5 7 2
+ 1 2 2 3
---------
```

(3)
```
  7 4 2 5
+ 2 3 5 3
---------
```

(4)
```
  4 6 5 3
+ 4 2 3 3
---------
```

(5)
```
  7 2 4 5
+ 2 4 3 3
---------
```

(6)
```
  2 8 4 7
+ 3 1 5 2
---------
```

(7)
```
  3 6 0 4
  4 2 4 2
+ 0 1 2 3
---------
```

(8)
```
  7 4 6 4
  1 0 1 2
+ 1 4 2 1
---------
```

(9)
```
  5 0 9 3
  2 7 0 4
+ 1 1 0 2
---------
```

(B) Solve the following:

(1) 2215 + 1322 = ______

(2) 4325 + 3210 = ______

(3) 4625 + 1234 = ______

(4) 3624 + 4172 = ______

(5) 7383 + 2514 = ______

(6) 4502 + 1224 + 4113 = ______

(7) 1102 + 1323 + 1340 = ______

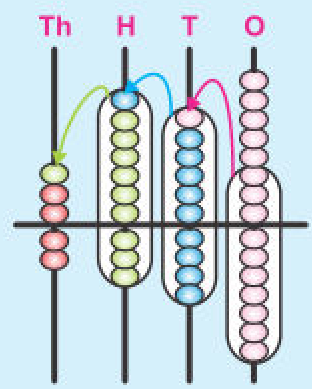
Now we add numbers up to 4-digits with carrying.

Example 1: Add 2658 and 2347

Starter: Using Example 1, the teacher will explain step wise, the vertical method of adding 4 digit numbers with carrying.
To check for student understanding, another sum will be written on the board and individual children will be asked to come to the board and solve the sum step by step.

Solution:

	Th	H	T	O
	1	1	1	
	2	6	5	8
+	2	3	4	7
	5	0	0	5



Step 1: Add ones
8 + 7 = 15 ones.
1 ten and 5 ones. Write 5 below ones and carry 1 ten to tens column.

Step 2: Add tens.
1 + 5 + 4 = 10 tens.
Write 0 below tens and carry 1 to hundreds column

Step 3: Add hundreds
1 + 6 + 3 = 10 hundreds.
Write 0 below hundreds and carry to thousands column.

Step 4: Add thousands
1 + 2 + 2 = 5 thousands
Write 5 below the thousands column.

or 2658 + 2347 = $\boxed{5005}$

Teacher's Note Teacher should explain the steps of addition of numbers with carrying up to 4-digit.

Example 2: **Add the following:**

3458, 1322, 2687

Th	H	T	O
1 3	1 4	1 5	8
1	3	2	2
+ 2	6	8	7
7	4	6	7

Step 1: Add ones
$8 + 2 + 7 = \boxed{17}$

Step 2: Add tens
$1 + 5 + 2 + 8 = \boxed{16}$

Step 3: Add hundreds
$1 + 4 + 3 + 6 = \boxed{14}$

Step 4: Add thousands
$1 + 3 + 1 + 2 = \boxed{7}$

or **3458 + 1322 + 2687 = $\boxed{7467}$**

EXERCISE 10

(A) Solve:

(1)
```
  4 6 2 6
+ 3 3 9 4
---------
```

(2)
```
  3 0 9 7
+ 4 9 7 4
---------
```

(3)
```
  7 3 8 5
+ 2 7 9 8
---------
```

(4)
```
  3 4 8 7
    9 3 2
+ 1 4 7 2
---------
```

(5)
```
  5 7 0 5
  2 8 9 9
+   1 0 2
---------
```

(6)
```
  3 7 8 9
  2 6 6 8
+   2 0 2
---------
```

(B) Add the following:

(1) **4866** and **2154**
(2) **3239** and **2453**
(3) **8370, 3745** and **235**
(4) **2555, 3876** and **1000**

(C) Solve the following:

(1) **3994 + 1089 =** ☐
(2) **3456 + 1268 =** ☐
(3) **2658 + 3846 =** ☐
(4) **3477 + 2955 =** ☐

Add numbers up to 100 using mental calculation strategies

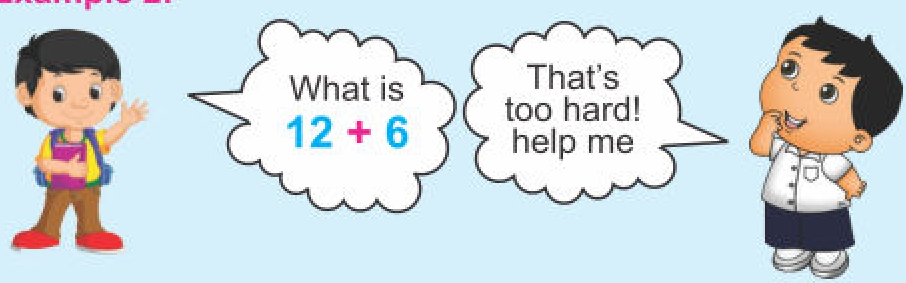
Starter: The teacher will explain a variety of mental strategies using the examples given.

Always remember ones are added to ones and tens are added to tens.

Example 1: Add mentally.

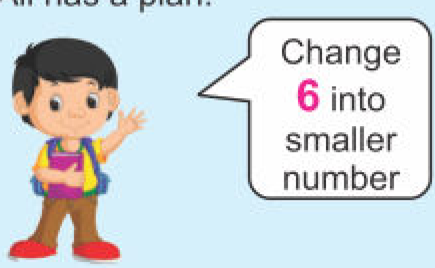
10 + 30	20 + 25
= 10 + 10 + 10 + 10	= 10 + 10 + 10 + 10 + 5
= 40	**= 45**

Example 2:



Can you show way in which Ahmed can make her sum easier?

Ali has a plan.



12 + 6
12 + 2 + 2 + 2
Write the sum
12 + 2 = 14
14 + 2 = 16
So, 16 + 2 = 18

Ahmed has another plan.

12 + 6
or 12 + 3 + 3
12 + 3 = 15
So, 15 + 3 = 18

Teacher's Note Teacher may tell the students any other way for mental addition which he/she thinks to be easier.

EXERCISE 11

(A) Add the following numbers by using more than one strategies.

(1) 14 + 5 (2) 15 + 6

(3) 16 + 8 (4) 21 + 11

(5) 22 + 6 (6) 11 + 32

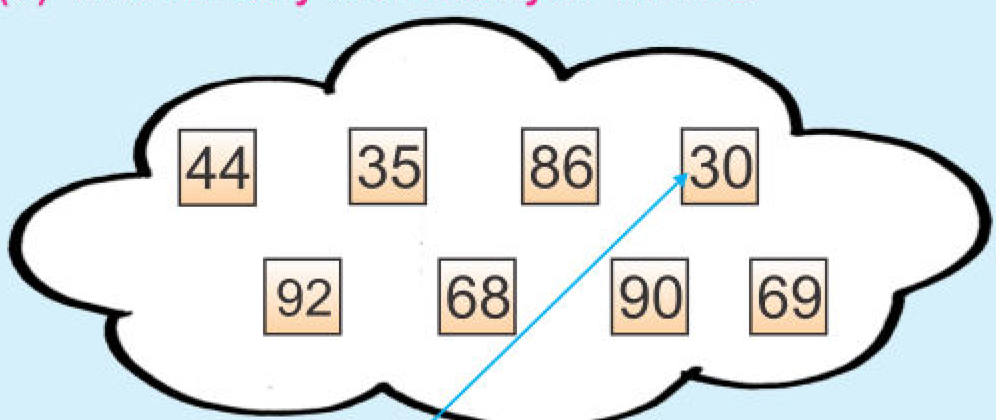
(B) Solve mentally.

(1) 25 + 20 = ☐ (2) 22 + 21 = ☐

(3) 46 + 14 = ☐ (4) 32 + 4 = ☐

(5) 42 + 28 = ☐ (6) 54 + 26 = ☐

(C) Add mentally and match your answer.



(1) 10 + 20 = 30 (2) 45 + 23 = ☐

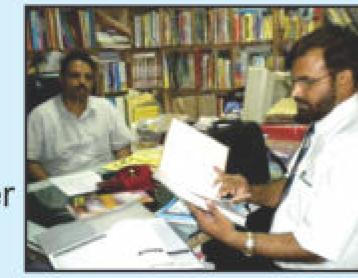
(3) 23 + 12 = ☐ (4) 22 + 22 = ☐

(5) 10 + 50 + 30 = ☐ (6) 18 + 38 + 36 = ☐

(7) 8 + 46 + 32 = ☐ (8) 20 + 13 + 36 = ☐

Solve real life problems involving addition

Example 1:



Aslam sells **3245** note books of Mathematics and **2568** note books of Sindhi in a month. Find the total number of note books he sold in that month.

Solution:

Number of Mathematics note books sold	=	3 ^{1}2 ^{1}4 5
Number of Sindhi note books sold	=	+ 2 5 6 8
Total number of note books sold	=	**5 8 1 3**

EXERCISE 12

(1) Rehana bought two books. One book has 328 pages and other has 1034 pages. How many pages in both?

(2) Aashir walks 800m in the morning and 1400m in the evening. How far does he walk altogether?

(3) In a village there are 3321 men, 2843 women and 1822 children. What is the total population of the village?

(4) In a Zoo 3451 visitors visited on Sunday, 2342 on Monday. How many visitors visited in these two days?

(5) Azhar reads a story book. On Saturday, he reads 236 pages and on Sunday 142 pages. How many pages does he read in both days.

(6) In a primary school there are 436 girls and 341 boys. How many students are altogether?

(7) In a city 4798 soft drinks sold on Friday and 3202 on Saturday. How many soft drinks were sold on both days?

Teacher's Note Teacher should give more examples of real life and practice to the students for making such sums.

2.2 SUBTRACTION

We have learnt to subtract numbers up to 3-digits in previous class. Let us revise.

EXERCISE 13

Solve.

(1) $\begin{array}{r} 88 \\ -\ 47 \\ \hline \end{array}$

(2) $\begin{array}{r} 76 \\ -\ 54 \\ \hline \end{array}$

(3) $\begin{array}{r} 639 \\ -\ 38 \\ \hline \end{array}$

(4) $\begin{array}{r} 473 \\ -\ 151 \\ \hline \end{array}$

(5) $\begin{array}{r} 378 \\ -\ 56 \\ \hline \end{array}$

(6) $\begin{array}{r} 297 \\ -\ 74 \\ \hline \end{array}$

(7) $\begin{array}{r} 356 \\ -\ 178 \\ \hline \end{array}$

(8) $\begin{array}{r} 625 \\ -\ 478 \\ \hline \end{array}$

(9) $\begin{array}{r} 243 \\ -\ 184 \\ \hline \end{array}$

(10) $\begin{array}{r} 702 \\ -\ 189 \\ \hline \end{array}$

(11) $\begin{array}{r} 958 \\ -\ 199 \\ \hline \end{array}$

(12) $\begin{array}{r} 342 \\ -\ 155 \\ \hline \end{array}$

First we subtract 4-digit numbers without borrowing.

Example: **Solve 6554 – 2313**

Solution:

	Th	H	T	O
	6	5	5	4
–	2	3	1	3
	4	2	4	1

Step 1: Subtract ones.
$4 - 3 = 1$
write 1 below ones column.

Step 2: Subtract tens.
$5 - 1 = 4$
write 4 below tens column.

Step 3: Subtract hundred
$5 - 3 = 2$
write 2 below hundreds column.

Step 4: Now subtract thousands.
$6 - 2 = 4$
write 4 below thousands column.

or **6554 – 2313**

Solution: **6554 – 2313 = 4241**

Note: Start from ones column first, then tens, hundred and last of all thousands.

Teacher's Note Teacher should help students to explain the steps of subtraction of numbers up to 4-digits.

EXERCISE 14

(A) Solve:

(1)
```
  4 2 6 9
- 3 0 3 2
---------
```

(2)
```
  9 9 6 8
- 2 7 3 4
---------
```

(3)
```
  7 6 9 5
- 3 2 5 3
---------
```

(4)
```
  8 7 5 6
- 3 2 3 4
---------
```

(5)
```
  6 5 9 8
- 4 3 8 6
---------
```

(6)
```
  6 9 7 6
- 2 3 4 3
---------
```

(7)
```
  5 7 5 4
- 2 3 5 2
---------
```

(8)
```
  3 8 7 9
- 2 3 4 6
---------
```

(9)
```
  4 5 8 9
- 3 4 6 6
---------
```

(B) Solve:

(1) 3455 – 1343 = ☐

(2) 6657 – 3246 = ☐

(3) 3689 – 1345 = ☐

(4) 8672 – 4350 = ☐

(5) 9563 – 5341 = ☐

Example: **Subtract 1446 from 3324**

	Th	H	T	O
	2	2	11	14
	~~3~~	~~3~~	~~2~~	~~4~~
–	1	4	4	6
	1	8	7	8

Step 1: 4 cannot be subtracted from 6.
So, we will take 1 tens from 2 tens.
As 1 ten = 10 ones. 10 + 4 = 14 ones
Then subtract 6 from 14 and write 8 at ones place.

Step 2: After taking 1 ten from 2 tens, 1 ten is left.
Since 4 cannot be taken away from 1 tens.
So, we will take 1 hundred from 3 hundreds.
1 hundred = 10 tens. 10 tens + 1 ten = 11 tens.
Then subtract 4 from 11 and write 7 at tens place.

Step 3: After taking 1 hundred from 3 hundred,
2 hundred left.
Since 4 cannot be taken away from 3 hundred.
So, we will take 1 thousand from 3 thousands.
1 thousand = 10 hundred and
10 hundreds + 2 hundred = 12 hundreds.
Thus, 12 – 4 = 8
Now we write 8 at hundred place.

Step 4: At last after taken 1 thousand from
3 thousands, left 2 thousands. Now subtract
1 from 2 and write 1 in thousands column.

or horizontally we write it as

3324 – 1446 = $\boxed{1878}$

Teacher's Note Teacher should explain the steps clearly and also give the sums horizontally and vertically.

EXERCISE 15

(A) Solve:

(1)
$$\begin{array}{r} {\scriptstyle 6\ 10\ 14} \\ 5\ \cancel{7}\ \cancel{1}\ \cancel{4} \\ -\ 3\ 5\ 4\ 6 \\ \hline \\ \hline \end{array}$$

(2)
$$\begin{array}{r} 6\ 6\ 0\ 8 \\ -\ 3\ 2\ 4\ 7 \\ \hline \\ \hline \end{array}$$

(3)
$$\begin{array}{r} 5\ 4\ 2\ 3 \\ -\ 3\ 9\ 8\ 7 \\ \hline \\ \hline \end{array}$$

(4)
$$\begin{array}{r} 3\ 9\ 4\ 0 \\ -\ 2\ 7\ 3\ 5 \\ \hline \\ \hline \end{array}$$

(5)
$$\begin{array}{r} 5\ 7\ 0\ 5 \\ -\ 2\ 9\ 3\ 4 \\ \hline \\ \hline \end{array}$$

(6)
$$\begin{array}{r} 8\ 0\ 0\ 4 \\ -\ 4\ 0\ 3\ 5 \\ \hline \\ \hline \end{array}$$

(7)
$$\begin{array}{r} 9\ 0\ 1\ 2 \\ -\ 7\ 6\ 5\ 4 \\ \hline \\ \hline \end{array}$$

(8)
$$\begin{array}{r} 3\ 4\ 5\ 2 \\ -\ 2\ 3\ 2\ 8 \\ \hline \\ \hline \end{array}$$

(9)
$$\begin{array}{r} 3\ 3\ 8\ 1 \\ -\ 2\ 6\ 4\ 5 \\ \hline \\ \hline \end{array}$$

(B) Complete the following:

(1) $6572 - 1459 = \boxed{}$ (2) $8095 - 7251 = \boxed{}$

(3) $4562 - 3571 = \boxed{}$ (4) $6559 - 2382 = \boxed{}$

(C) Subtract horizontally:

(1) **3254** from **7293** (2) **4972** from **6743**

(3) **2484** from **6512** (4) **5328** from **8672**

Subtract numbers up to 100 using mental calculation strategies

Always remember ones are subtracted from ones and tens are subtracted from tens.

Example:

Subtract the following by using more than one strategies.

(i) **53 – 8**
= 53 – 3 – 5
= 50 – 5 = 45
Or 53 – 8 = 50 + 3 – 3 – 5
= 50 – 5
= 45

(ii) **72 – 6**
= 72 – 2 – 4
= 70 – 4 = 66
Or 72 – 6 = 70 + 2 – 1 – 5
= 70 – 5 + 2 – 1
= 65 + 1 **= 66**

Activity **Subtract.**

(1) $\boxed{50} \xrightarrow{-5} \boxed{} \xrightarrow{-10} \boxed{} \xrightarrow{-2} \boxed{}$

(2) $\boxed{100} \xrightarrow{-30} \boxed{} \xrightarrow{-5} \boxed{} \xrightarrow{-20} \boxed{}$

EXERCISE 16

1. **Subtract the following numbers by using more than one strategies.**

(i) 15 – 6	(ii) 70 – 8	(iii) 78 – 12
(iv) 89 – 26	(v) 56 – 35	(vi) 93 – 27

Teacher's Note Teacher may tell the students any other way for mental calculation strategies of subtraction.

2. Solve.

(i) $47 - 16 = \square$
(ii) $19 - 13 = \square$
(iii) $50 - 12 = \square$
(iv) $46 - 21 = \square$
(v) $89 - 40 = \square$
(vi) $67 - 14 = \square$
(vii) $97 - 18 = \square$
(viii) $100 - 87 = \square$

3. Solve the following mentally:

(i) There are **9** birds are sitting on a tree. **3** flew away.
How many birds remain there?

(ii) Abdul Raheem has **10** sheeps and he has **5** cows.
How many more sheeps than cows does he has?

(iii) Sania has **42** kg of rice.
She gave **20** kg to her sister.
How much rice is left with her?

Solve real life problems involving subtraction

Example:

Nawaz had **1245** packets of chocolates. He sold **1089** packets. How many packets left?

Solution:

He has	$1\ \cancel{2}^{1}\ \cancel{4}^{13}\ {}^{1}5$	packets
He sold	$-1\ 0\ 8\ 9$	packets
	$0\ 1\ 5\ 6$	packets left

EXERCISE 17

(1) A shopkeeper has **850** chocolates of milk. He sold **480** chocolates. How many chocolates are left?

(2) Ali has two notes in his wallet. Their value together is **Rs 600.** One note is of **Rs 100**. What is the value of other note?

(3) A factory manufacturers **2460** cars in a year. **1946** cars are sold. How many cars are yet to be sold?

(4) Shamim collected **Rs 4950** as donation for flood affected. She purchased grocery and utensils **Rs 2895** for the affected villagers. How much amount is still left with her?

(5) A bakery had **4500** packet of cookies. **1862** were sold. How many were left?

(6) The population of a village comprises **9015** adult persons. **4386** are men, find the number of women.

2.3 MULTIPLICATION

Use the term 'product' for multiplication of two numbers

Starter : The teacher will draw 3 starfish on the board. Each starfish should have 5 arms. The teacher will ask the children how many arms are there altogether? The teacher will explain that one method of finding this out is to do this: 5 +5 +5 =15

Which means, 3 times 5 or 3 by 5

Or 3 x 5 = 15

The teacher will further explain that when two or more numbers are multiplied together their total is called product

So the product of 3 and 5 is 15.

We have learnt multiplication in previous class.

There are 3 Starfish.
Each Starfish has 5 arms.
How many arms are altogether?

5 + 5 + 5 = 15

It means

3 times 5 ➡ 3 by 5

Or **3 x 5 = $\boxed{15}$**

The product of **3** and **5** is **15**.

Teacher's Note Teacher should revise the multiplication concept through process of repeating addition and use.

We can use the term **"product"** for **"multiplication"** of numbers.

Examples: Find the product of:

(1) **4** and **6**.

Solution:

6 by 4 means 6 times 4

$4 \times 6 = \boxed{24}$

The product of **4** and **6** is **24**.

(2) **8** and **5**.

Solution:

Product of 8 and 5 means 8 by 5

or 8 times 5

$8 \times 5 = \boxed{40}$

or

$$\begin{array}{r} 8 \\ \times\ 5 \\ \hline 40 \\ \hline \end{array}$$

Thus, the product is **40**.

EXERCISE 18

(A) Find the product of the following:

(1) $3 \times 4 = \boxed{12}$

(2) $2 \times 6 = \boxed{}$

(3) $3 \times 3 = \boxed{}$

(4) $7 \times 5 = \boxed{}$

(5) $5 \times 9 = \boxed{}$

(6) $8 \times 3 = \boxed{}$

(B) Solve the following:

(1)	The product of 5 and 4 is	5 x 4	=	20
(2)	The product of 2 and 8 is		=	
(3)	The product of 5 and 6 is		=	

Develop multiplication tables for 6, 7, 8 and 9

We have learnt tables of 2, 3, 4, 5 and 10 in previous class.
Now we develop table of 6.

Addition table	Way of reading	Multiplication table
6	1 six is 6	1 x 6 = **6**
6 + 6	2 sixes are 12	2 x 6 = **12**
6 + 6 + 6	3 sixes are 18	3 x 6 = **18**
6 + 6 + 6 + 6	4 sixes are 24	4 x 6 = **24**
6 + 6 + 6 + 6 + 6	5 sixes are 30	5 x 6 = **30**
6 + 6 + 6 + 6 + 6 + 6	6 sixes are 36	6 x 6 = **36**
6 + 6 + 6 + 6 + 6 + 6 + 6	7 sixes are 42	7 x 6 = **42**
6 + 6 + 6 + 6 + 6 + 6 + 6 + 6	8 sixes are 48	8 x 6 = **48**
6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6	9 sixes are 54	9 x 6 = **54**
6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6 + 6	10 sixes are 60	10 x 6 = **60**

Activity **Count in 6s and fill the missing numbers.**

6, ________ **,** ________ **, 24,** ________ **,** ________

42, ________ **,** ________ **,** ________ **, 66,** ________

Teacher's Note Teacher should help the students to develop table of 6, 7, 8 and 9 by using concrete material.

Table of 7.

Addition table	Way of reading	Multiplication table
7	1 seven is 7	1 x 7 = 7
7 + 7	2 sevens are 14	2 x 7 = 14
7 + 7 + 7	3 sevens are 21	3 x 7 = 21
7 + 7 + 7 + 7	4 sevens are 28	4 x 7 = 28
7 + 7 + 7 + 7 + 7	5 sevens are 35	5 x 7 = 35
7 + 7 + 7 + 7 + 7 + 7	6 sevens are 42	6 x 7 = 42
7 + 7 + 7 + 7 + 7 + 7 + 7	7 sevens are 49	7 x 7 = 49
7 + 7 + 7 + 7 + 7 + 7 + 7 + 7	8 sevens are 56	8 x 7 = 56
7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7	9 sevens are 63	9 x 7 = 63
7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7 + 7	10 sevens are 70	10 x 7 = 70

Activity Count in 7s and fill in the blanks.

(1) 2 x 7 = 14 (2) 3 x 7 = ____

(3) 6 x 7 = ____ (4) 8 x 7 = ____

(5) 9 x 7 = ____ (6) 7 x 7 = ____

Table of 8.

Addition table	Way of reading	Multiplication table
8	1 eight is 8	1 x 8 = 8
8 + 8	2 eights are 16	2 x 8 = 16
8 + 8 + 8	3 eights are 24	3 x 8 = 24
8 + 8 + 8 + 8	4 eights are 32	4 x 8 = 32
8 + 8 + 8 + 8 + 8	5 eights are 40	5 x 8 = 40
8 + 8 + 8 + 8 + 8 + 8	6 eights are 48	6 x 8 = 48
8 + 8 + 8 + 8 + 8 + 8 + 8	7 eights are 56	7 x 8 = 56
8 + 8 + 8 + 8 + 8 + 8 + 8 + 8	8 eights are 64	8 x 8 = 64
8 + 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8	9 eights are 72	9 x 8 = 72
8 + 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8	10 eights are 80	10 x 8 = 80

Activity Count in 8 and complete.

1	2	3	4	5	6	7	8	9	10
x8	16								

Table of 9.

Addition table	Way of reading	Multiplication table
9	1 nine is 9	1 x 9 = **9**
9 + 9	2 nines are 18	2 x 9 = **18**
9 + 9 + 9	3 nines are 27	3 x 9 = **27**
9 + 9 + 9 + 9	4 nines are 36	4 x 9 = **36**
9 + 9 + 9 + 9 + 9	5 nines are 45	5 x 9 = **45**
9 + 9 + 9 + 9 + 9 + 9	6 nines are 54	6 x 9 = **54**
9 + 9 + 9 + 9 + 9 + 9 + 9	7 nines are 63	7 x 9 = **63**
9 + 9 + 9 + 9 + 9 + 9 + 9 + 9	8 nines are 72	8 x 9 = **72**
9 + 9 + 9 + 9 + 9 + 9 + 9 + 9 + 9	9 nines are 81	9 x 9 = **81**
9 + 9 + 9 + 9 + 9 + 9 + 9 + 9 + 9 + 9	10 nines are 90	10 x 9 = **90**

Activity Count in 9 and complete.

1	2	3	4	5	6	7	8	9	10
x9									

Multiply 2-digit numbers by 1-digit number

Example: There are 3 bags and each bag has 12 toffees.

It means 3 times 12 *or* 3 by 12.

How many toffees in all?

	T	O
	1	2
x		3
	3	6

Step 1: When we multiply the Ones we get 3 x 2 = 6. Write 6 below Ones column.

Step 2: Now we multiply the ten 3 x 1 = 3 Write 3 below tens column.

Example: **Solve:** **Multiply 75 by 4.**

	T	O
	2 7	5
x		4
3	0	0

Step 1: 4 x 5 = 20 ones
= 2 tens

Step 2: 4 x 7 = 28 tens
Now 28 + 2 = 30 tens

Teacher's Note

On the board, the teacher will draw three bags, each containing 12 toffees. Ask the children how many toffees are there in all?
Teacher should have the students to understand the process of multiplication of 2-digit numbers by 1-digit number.

EXERCISE 19

(A) Solve the following:

(1)
$$\begin{array}{r} 14 \\ \times\ 2 \\ \hline \\ \hline \end{array}$$

(2)
$$\begin{array}{r} 31 \\ \times\ 2 \\ \hline \\ \hline \end{array}$$

(3)
$$\begin{array}{r} 21 \\ \times\ 4 \\ \hline \\ \hline \end{array}$$

(4)
$$\begin{array}{r} 12 \\ \times\ 4 \\ \hline \\ \hline \end{array}$$

(5)
$$\begin{array}{r} 23 \\ \times\ 3 \\ \hline \\ \hline \end{array}$$

(6)
$$\begin{array}{r} 35 \\ \times\ 2 \\ \hline \\ \hline \end{array}$$

(7)
$$\begin{array}{r} 11 \\ \times\ 6 \\ \hline \\ \hline \end{array}$$

(8)
$$\begin{array}{r} 53 \\ \times\ 6 \\ \hline \\ \hline \end{array}$$

(9)
$$\begin{array}{r} 83 \\ \times\ 4 \\ \hline \\ \hline \end{array}$$

(B) Multiply:

(1) 13 by 2

(2) 64 by 4

(3) 23 by 5

(4) 22 by 6

(5) 21 by 7

(6) 15 by 3

(7) 37 by 5

(8) 14 by 6

Multiply a number by zero

Alia has 0 apples.

Kareem has 0 apples.

Kamran has 0 apples.

$0 + 0 + 0 = 0$

Or $3 \times 0 = \boxed{}$

Zero multiplied by any number is always Zero.

EXERCISE 20

Solve:

(1) $0 \times 2 = \boxed{}$	(2) $0 \times 6 = \boxed{}$
(3) $5 \times 0 = \boxed{}$	(4) $10 \times 0 = \boxed{}$
(5) $4 \times 0 = \boxed{}$	(6) $0 \times 15 = \boxed{}$
(7) $7 \times 0 = \boxed{}$	(8) $1 \times 0 = \boxed{}$
(9) $3 \times 0 = \boxed{}$	(10) $8 \times 0 = \boxed{}$
(11) $0 \times 9 = \boxed{}$	(12) $0 \times 13 = \boxed{}$

Teacher's Note Teacher should give the concept of multiplication of any number by zero is always zero (0).

Apply mental mathematical strategies to multiply numbers up to the table of 10.

Example 1: **Multiply the following by using the mathematical strategies.**

(i) 3×5

Solution:

$3 \times 5 = (5 - 2) \times 5$

$= 5 \times 5 - 2 \times 5$

$= 25 - 10$

$= 15$

(ii) 12×13

Solution:

$12 \times 13 = (10 + 2) \times (10 + 3)$

$= 10 \times (10 + 3) + 2 \times (10 + 3)$

$= 10 \times 10 + 10 \times 3 + 2 \times 10 + 2 \times 3$

$= 100 + 30 + 20 + 6$

$= 150 + 6 = 156$

EXERCISE 21

Multiply the following by using mathematical strategies.

(1) 4 x 5	(2) 6 x 5	(3) 7 x 6
(4) 9 x 6	(5) 9 x 12	(6) 8 x 14
(7) 11 x 13	(8) 14 x 16	(9) 23 x 27
(10) 29 x 32	(11) 35 x 40	(12) 65 x 75
(13) 45 x 27	(14) 68 x 14	(15) 79 x 21
(16) 85 x 35	(17) 92 x 25	(18) 95 x 40

Solve real life problems involving multiplication of 2-digit numbers by 1-digit numbers

Example: **25** glasses of water fill **1** bucket. How many glasses of water needed to fill 6 such buckets?

$$\begin{array}{r} {}^{3}2\ 5 \\ \times\ \ 6 \\ \hline 1\,5\,0 \\ \hline \end{array}$$

Step 1: 6 x 5 = 30 ones

$\boxed{= 3 \text{ tens } 0 \text{ ones}}$

Step 2: 6 x 2 = 12 tens

Add 3 tens

12 tens + 3 tens = 15 tens

Total number of glasses = 150

EXERCISE 22

(1) In a parking area there is capacity of **15** motor bikes in a row. How many motor bikes are there in such **6** rows?

(2) There are **7** days in a week. How many days are there in **40** weeks?

(3) There are **8** trees in the park. **6** birds are sitting on each tree. How many birds are there in all?

(4) A packet contains **54** chocolates. How many chocolates are there in **7** such packets?

(5) There are **3** scoops of ice cream in a cup. How many scoops of ice cream used to make **48** cups of ice creams?

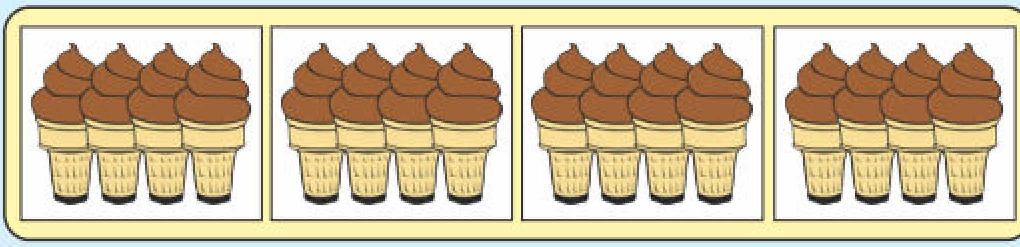
(6) There is space of **25** books in one shelf. How many books can be kept such **6** shelves?

2.4 DIVISION

Divide 2-digit numbers by 1-digit numbers (with zero remainder).

Starter: On the blackboard the teacher will draw four boxes, each containing four ice cream cones.

Example 1: There are 16 cones in a box of ice cream with 4 in each set. How many 4's are there in 16?



16 – 4 = 12
12 – 4 = 8
8 – 4 = 4
4 – 4 = 0

Or we can solve it by using division

$$\begin{array}{r} 4 \\ 4\overline{)16} \\ -16 \\ \hline 00 \end{array}$$

4 is contained 4 times in 16.

So, $16 \div 4 = \boxed{4}$

Process of division

$$\begin{array}{r} 5 \\ 6\overline{)30} \\ -30 \\ \hline 0 \end{array}$$

Quotient

Divisor

Dividend

Remainder

Teacher's Note Teacher should explain the long division method of division with zero remainder.